

Geo Flood Agent

Natural language → GIS code → isolated
execution → cascade analysis

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The problem: what hurts is the facility that is NOT under water but stops

Hazard maps only tell you where the water goes.

A hospital on high ground still goes dark if its substation sits in the lowlands.

That cascade never shows up in raster analysis — you have to walk the relationships.

"If the lower Arakawa rises 3 m, which facilities are affected?"

One question triggers GIS code generation → isolated execution → graph cascade analysis → briefing, returned in a form a municipal emergency operations center can act on.

Cascade example

Ueno Plateau Hospital

elevation 19 m · not flooded

↓ **POWERED_BY** ↓

Kita-Senju Substation

elevation -0.4 m · 3.4 m under water

→ **hospital loses power while staying dry**

Architecture: one role per platform that would not work without it

NOSANA

LLM inference (decentralized GPU)

- Writes the GIS script from the question
- OpenAI-compatible endpoint, one-line swap
- Turns results into an EOC briefing

DAYTONA

Isolated sandbox

- Safely runs arbitrary LLM-written code
- Fetches GSI elevation tiles → flood mask → map PNG
- create → upload → exec → download → delete

NEO4J

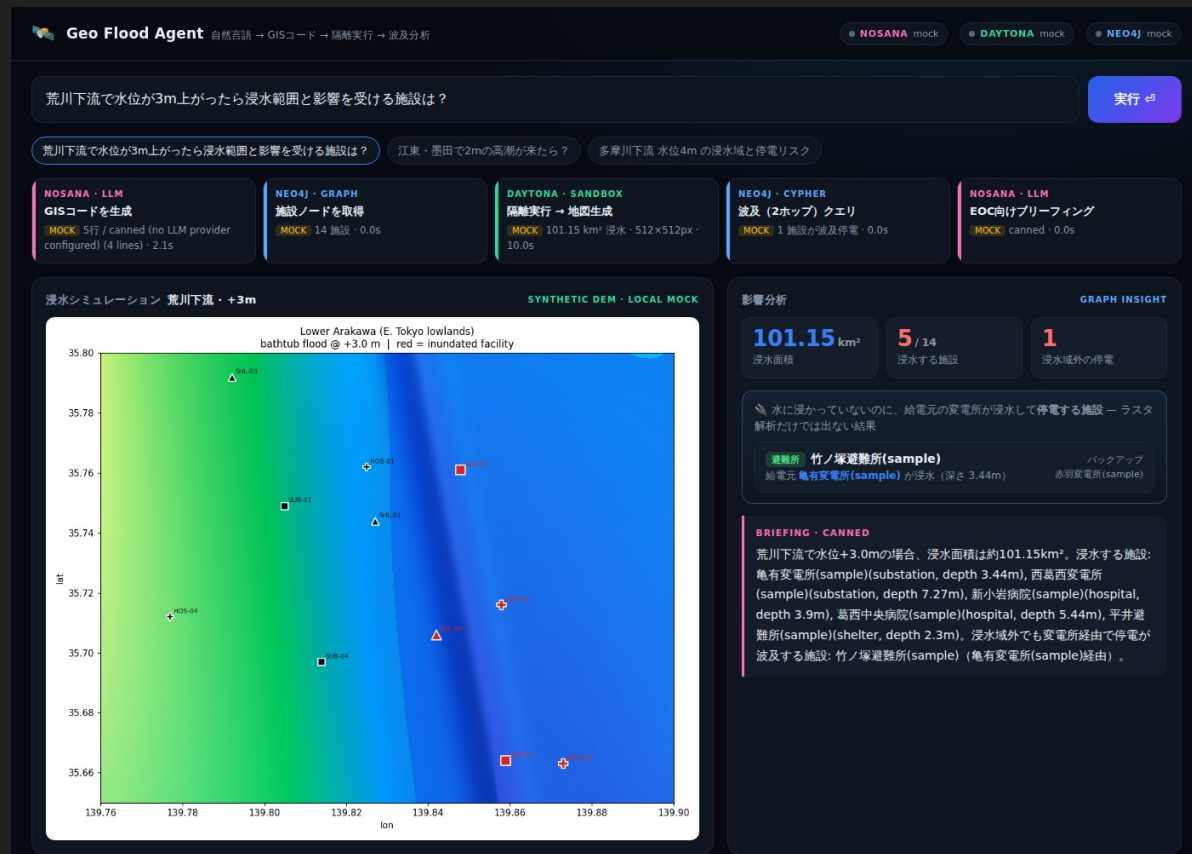
Knowledge graph

- Facilities and power feeds as a graph
- Writes each scenario as FLOODED_IN edges
- 2-hop Cypher finds outages outside the water

Question → [Nosana] code → [Neo4j] facilities → [Daytona] execute → [Neo4j] cascade query → [Nosana] briefing

Every call is flag-switchable; a platform that is down falls back to a same-interface mock, labelled MOCK in the UI

Demo: one question → flood map and the power cascade



101 km²

flooded area (+3.0 m, lower Arakawa)

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facilities under water

1

facility losing power outside the water

Each pipeline stage shows LIVE or MOCK. Generated code, Cypher and the live log are all exposed in the UI.

Where we are, and what comes next

Honest status today

- Neo4j Aura: live — scenario writes and the 2-hop cascade query
- Daytona: sign-up from the venue was blocked by anomaly detection for our whole team (confirmed with support). The same script runs through the local fallback; the integration is 30 lines in agent/sandbox.py and goes live by pasting an API key
- Nosana: OpenAI-compatible client implemented; no endpoint yet, so a canned script stands in
- Elevation is real GSI tile data; facilities and power feeds are sample data

Next

- Run SAR flood segmentation on Nosana GPUs and replace the bathtub model with observations
- Daytona snapshots for 1-second starts; evaluate water levels 1–5 m as N parallel sandboxes
- Load real municipal infrastructure (grid, shelters) into Neo4j and operate it in the EOC
- github.com/0xtkey256/geoflood